

**NAME: TANKISO MASOEBE**

## **COMPUTER ARCHITECTURE ADVANCED wk5**

### **1.SCENARIO ANALYSIS:**

A superscalar web server uses a processor that can execute multiple instructions at the same time during a single clock cycle. For example, when many users access a website simultaneously, the server receives different requests such as loading web pages, retrieving database information, processing user logins, and sending files. A superscalar processor improves performance by allowing several independent instructions from these tasks to be executed in parallel using multiple execution units. For instance, while one instruction accesses data from memory, another can perform calculations and another can process a different user request. This reduces the time required to complete tasks and allows the web server to handle more users efficiently. Superscalar processing is especially useful for busy websites and online services where many operations must be performed quickly. By executing multiple instructions simultaneously, the processor increases throughput, improves response time, and makes better use of available CPU resources. Therefore, superscalar architecture helps web servers provide faster and more reliable services.

### **2.CONCEPT RESEARCH:**

Dynamic scheduling is a CPU technique that allows instructions to be executed when their required resources and data are available rather than strictly following the original program order. It helps processors improve performance by reducing delays caused by dependencies between instructions. When one instruction is waiting for data from memory, the processor can execute another independent instruction instead of remaining idle. Dynamic scheduling uses hardware components to monitor instruction dependencies and determine the best order for execution. This technique increases instruction-level parallelism and makes better use of the processor's execution units. It is commonly used in modern superscalar processors to improve speed, efficiency, and overall system performance while maintaining correct program results.

### **3.TOOL PRACTICE:**

A superscalar pipeline can be designed with several instructions being processed simultaneously at different stages. A simple five-stage pipeline consists of Fetch (IF), Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). In a superscalar processor, multiple instructions can enter the pipeline during the same clock cycle. For example, Instruction 1 can be in the Execute stage while Instruction 2 is being Decoded and Instruction 3 is being Fetched. I designed the pipeline with two parallel execution paths to allow two instructions to be processed at once. This exercise helped me understand how superscalar processors increase instruction throughput and

reduce execution time. It also showed me the importance of handling instruction dependencies correctly.

#### 4.DiAGRAM:

The superscalar diagram illustrates how a processor can execute multiple instructions simultaneously by using parallel pipelines. The diagram begins with the Fetch stage, where multiple instructions are retrieved from memory. During the Decode stage, the instructions are analyzed and prepared for execution. They then move through different execution units, allowing independent instructions to be processed at the same time. After execution, the results pass through the Memory stage and finally reach the Write Back stage. The diagram shows that several instructions can occupy different pipeline stages during the same clock cycle. This improves instruction throughput, reduces processing time, and increases overall CPU performance.

